KINGDOM AI OS

Master Architecture Bible & Build Specification

24 Workspaces • Offline-First • Uncensored Priority • Modular Forever

Version 1.0 — Intake Complete
September 30, 2026

Source of Truth for Manus 1.6 implementation.
Financial Platform remains architecturally separate.

1. Executive Overview & Core Decisions

Kingdom AI OS is not a chatbot with plugins. It is a modular AI operating system whose models, tools, agents, and media engines can be swapped without rebuilding the core. The design prioritizes offline capability, uncensored/low-guardrail models for sovereignty and research work, character consistency for realistic diverse humans, and a never-break plugin architecture.

Foundational Decisions

- Do not code the entire system from a raw name list. Sequence: Verified inventory → Compatibility matrix → Architecture → Automated installer → Services → UI → Agents → Tests → Deployment.
- Open WebUI is a strong UI candidate but not irreplaceable. Use v0.6.5 (last fully BSD-3) or earlier as base; build a custom Kingdom frontend/control plane so it can be replaced later.
- Model registry owns the models. Inference engine (vLLM, llama.cpp, Ollama, SGLang, Colibrì, MLX) is swappable.
- Colibrì is a first-class local engine for huge MoE models (GLM-5.x 744B, Kimi K3 ~2.8T, DeepSeek V4 Flash, etc.) by streaming experts from NVMe.
- Financial Platform is completely separate with controlled API only. No browser agent ever receives raw ledger or credential access.
- Simplified routing: OpenRouter (online) + one intelligent layer (optional GodMode-style) + Local (Colibrì + GPU/CPU). Avoid stacking many routers.
- Every component gets a registry record with license, offline status, hardware, security profile, alternatives, and verification status.

2. Locked Top-Level Tab Structure (24 Workspaces)

Tabs are workspaces. Open-source projects, models, and agents live inside them. Adding 50 more OSINT tools or 100 models does not create new tabs.

#	Tab	Primary Contents
1	■ Home / Command Center	Global chat, dashboard, notifications, agent/system status, quick launch
2	■ LLM / Models	Full registry, local/API/free/uncensored pools, router, health, benchmarks
3	■ Research	Deep/web/academic/news/document research, citations, research agents
4	■■ Legal	Cases, claims, litigation, sovereignty, treaties, evidence, legal docs
5	■■ OSINT	Investigation tools, sources, threat intel, exposure monitoring
6	■ Finance	Analysis, modeling, treasury interfaces (no direct ledger)
7	■ Investments	Portfolio, markets, SPVs, valuation, research
8	■ Real Estate	Properties, research, maps, valuation, documents
9	■ Business / CRM	Companies, contacts, pipelines, operations
10	■ Leads	Generation, enrichment, prospecting, outreach
11	■■ City / Urban Planning	Master planning, architecture, infrastructure, simulations
12	■■ Engineering	CAD/tech, robotics, simulation, calculations
13	■ Coding / Development	Agents, IDEs, repos, app factory, debugging
14	■ Web / Scraping	Browser agents, crawlers, data acquisition, automation
15	■ News / Magazines	Aggregation, RSS, feeds, publishing
16	■ Social / Marketing	Publishing, scheduling, campaigns, analytics
17	■ Fashion	Design, styling, lookbooks, virtual models
18	■ Images / Visual AI	Generation, editing, OCR, Human Studio, ComfyUI/FLUX
19	■ Video / UGC / TV / Movies	Generation, editing, remake engine, characters, clipping
20	■ Music / Audio	Generation, stems, mixing, transcription, voice
21	■ Agents / Communications	Agent creator/library, phone/email/messaging, identities
22	■ Tasks	Global open / in-progress / completed / failed / scheduled
23	■ Open Source / Apps	Tool registry, Dify, MCP, plugins, app factory
24	■ Security / Admin	IAM, secrets, audit, health, backups, sandboxes, threat response

Internal Structure of Every Tab

Chat → Agents → Active/Closed Tasks → Knowledge → Files/Uploads → Tools → Models → Outputs → Automations → Settings. This consistency means the UX is recognizable whether you are in Legal, Video, or OSINT, while the underlying sandbox, tools, models, and permissions differ.

3. UI Specification (Based on Uploaded Interface)

Dark theme primary. Bright accent palette: orange, teal, pink, purple, green (avoid heavy pure blue). White text for content readability. Support optional light mode.

Layout

Left Sidebar (collapsible): Model selector at top → Search → Home → Chat (global) → Prompt Library → Integrations / Open Source → Pinned conversations (color-coded by workspace) → History (Today / Yesterday / 7 days).

Main Content: Welcome / Command Center with feature cards (Task Automation, Multi-language, Image Generation, Code, Research, Legal, Video, Music, etc.). Suggestion chips. Chat input bar with 'Ask me anything…', Saved prompts, Attach. Top-right: Export Chat, Status/Upgrade, Notifications bell.

Top or secondary navigation: The 24 workspace tabs (or collapsible groups). Clicking a tab switches the entire context into that sandbox while preserving global chat access.

Global overlays: Active Tasks drawer, Agent status panel, System health, Security alerts.

Color Coding

Each workspace has a signature bright color for task chips, history items, and status indicators so the user can instantly see which domain a task or conversation belongs to.

4. High-Level Architecture

Control Plane + Data Plane

```
KINGDOM AI OS
■
■■■ CONTROL PLANE
■   ■■■ Tasks / Agents / Models
■   ■■■ Model Router
■   ■■■ Credential Broker
■   ■■■ Audit / SIEM
■   ■■■ Sandbox Manager
■
■■■ DATA PLANE
■   ■■■ Files / Knowledge / RAG
■   ■■■ Character Registry (Human Studio)
■   ■■■ Workspace Databases
■
■■■ 24 WORKSPACES (isolated sandboxes)
    ■■■ Research / Legal / OSINT
    ■■■ Finance / Investments / Real Estate (interfaces only)
    ■■■ Media (Images / Video / Music / Fashion)
    ■■■ Coding / Engineering / City Planning
    ■■■ Agents / Tasks / Security / Open Source
```

Model Control Plane

```
                MODEL CONTROL PLANE
                       ■
     ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
     ■              ■              ■
   LOCAL          REMOTE         HYBRID
     ■              ■              ■
Colibrì        OpenRouter     Kaggle workers
llama.cpp      Free APIs      Token Router
vLLM / SGLang  NVIDIA/Bytez   etc.
Ollama / MLX
```

Deployment Topology

```
Internet → Cloudflare (DNS / WAF / Tunnel / DDoS protection)
                ↓
Local Control Plane (your server or laptop)
  - Persistent DB, auth, orchestration, files, containers
                ↓
Model Router
  ■■■ Local GPU/CPU + Colibrì (huge MoEs streamed from NVMe)
  ■■■ Kaggle (temporary replaceable GPU workers)
  ■■■ OpenRouter / free APIs (online fallback)
```

5. Model Registry — Priority & Verified Highlights

Every model receives a full registry record (id, repository, license, hardware estimates, modalities, guardrail class, offline status, primary/backup jobs, verification status). Below are the highest-priority and verified foundations. Community uncensored/obliterated variants are preferred for research/sovereignty when license and safety allow.

Colibrì-Supported Huge MoEs (Local Disk Streaming)

Colibrì (Apache-2.0, pure C) keeps dense layers in RAM (~10 GB int4) and streams routed experts from NVMe. Enables 744B–2.8T class models on consumer hardware with sufficient RAM + fast storage. Speed depends on disk.

Model Family	Approx Size	Notes
GLM-5.2 / 5.3	~744B total / ~40B active	Core Colibrì target; strong general + reasoning
GLM-5.3-Flash	~321B / vision	Multimodal, faster path
Inkling	~975B	Supported family
Kimi K3	~2.8T total	Largest currently supported
DeepSeek V4 Flash / V4.1 Flash	~284B–552B	Strong reasoning / coding candidates
Qwen3.x variants	Various	Including Flash-Next style

Other Priority Local / Hybrid Engines

- llama.cpp / KoboldCpp / MLC LLM — broad GGUF support, excellent offline.
- Ollama / LocalAI / Jan / LlamaFile — easy local serving.
- vLLM / SGLang / TensorRT-LLM — high-throughput when GPU available.
- OpenRouter — primary online gateway with free-tier models and health checks.
- Uncensored / low-guardrail community fine-tunes (Dolphin-style, Abliterated, etc.) — prioritized for research/legal/sovereignty tabs when verified.

Status note: Exact community variants (specific GLM-5.3 uncensored FP8, DeepSeek V4 Flash uncensored GGUF, Qwen 3.8 variants, etc.) are inventoried with NEEDS VERIFICATION until exact Hugging Face / GitHub URLs and licenses are confirmed.

6. Open-Source Tool Registry — By Category (Verified Strong Candidates First)

Tools are registered per workspace. Duplicates are kept with relationship notes rather than deleted. Suggestions for stronger alternatives appear where relevant.

■ Video / UGC / Remake Engine

- VACE (Wan2.1-VACE 1.3B & 14B) — All-in-one R2V / V2V / masked / Move/Swap/Reference/Expand/Animate. Apache-2.0 variants available. Primary recommendation.
- Wan2.1 family — Text-to-video, image-to-video, editing. Local inference paths exist.
- LTX / LTX-2 ecosystem — Image-to-video, keyframe, extension, video-to-video, audio sync, ComfyUI integration.
- Open-Sora — Open video generation baseline.
- ComfyUI + custom nodes — Universal media workflow engine.
- FFmpeg — Assembly, clipping, encoding (always present).
- Human / talking-head systems (MimicTalk / EchoMimic style, FaceFusion, DUiX-related where verified) for avatar/lip-sync.

■ Images / Human Studio / Character Consistency

- ComfyUI + FLUX / SD ecosystem + IP-Adapter implementations (XLabs etc.) for reference-conditioned generation.
- InvokeAI, Automatic1111/SD-WebUI, Diffusers — generation and editing baselines.
- Character ID system: permanent internal ID → face embedding + reference pack + body/hair/skin/clothing library → generation → identity + temporal verification.
- Explicit design requirement: realistic diverse skin tones (including deep brown / accurate melanin), varied body proportions, hair types (braids, locs, coils, curls, protective styles), anatomy (hands/feet), lighting consistency.
- OCR / Document visuals: MinerU (preferred multi-format + citations), PaddleOCR, Marker, Unstructured.

■ Document Intelligence & Research

- MinerU — Local multi-format (PDF, Office, images, etc.), quality tiers, stable page/block locators for citations. Strong RAG fit.
- PaddleOCR — Multilingual OCR + document VLM.
- Marker, Unstructured, Docling-style parsers as alternatives/fallbacks.
- SearXNG — Self-hosted metasearch (privacy-preserving).
- RSSHub + Folo — Feed aggregation.
- Crawl4AI, Firecrawl, Browser Use — Web acquisition.

■ Browser Agents & Web Automation

- Browser Use (MIT) — Leading open-source browser agent (local + optional hosted). Primary recommendation.
- Playwright + custom workers — Deterministic automation.
- open-browser-use style projects — Operate existing signed-in browser contexts.
- Critical security: Agents request permission → Credential Broker issues scoped secret → Action → Receipt → Audit. High-impact actions (purchases, account creation) require explicit approval.

■ Coding Agents & Development

- OpenHands — Docker-isolated coding agents that can modify code, run commands, browse, call APIs.
- Cline, Aider, Continue-style, custom Kingdom coding agents.
- Every coding agent receives a disposable workspace.
- App factory / Dify integration for building internal apps (including future trading/currency interfaces).

■ Music / Audio Studio

- Stem separation: Demucs (archived — keep for compatibility/reference only; prefer maintained forks or alternatives).

- StemDeck / StemForge style local tools for separation + mixing + MIDI.
- Whisper.cpp / Whisper WebUI — Transcription.
- F5-TTS, GPT-SoVITS, ChatTTS, Coqui-style TTS for voice.
- Full workstation: beat/melody/bass/drum generation, pitch correction, mixing, mastering, loudness, arrangement, export.

■ Security Stack

- Keycloak — Open-source IAM (auth, federation, fine-grained authorization). Apache-2.0.
- OpenBao / Vault-style — Secrets management.
- Wazuh — SIEM / XDR / endpoint monitoring.
- CrowdSec — Behavioral IDS/IPS/WAF/bot detection (monitor its own advisories).
- Suricata — Network IDS/IPS.
- Audit logging of every high-impact agent action. Continuous vulnerability monitoring of the security tools themselves.

■ Evaluation & Observability

- DeepEval — Task completion, relevance, hallucination metrics. Apache-2.0.
- Langfuse — Traces, prompts, datasets, agent workflows (self-hostable).
- Promptfoo — Model/prompt/RAG/agent evaluation + red-team. MIT.
- RAGAS, LM Eval Harness, Helicone-style as supporting tools.

Other Notable Categories (Inventory Preserved)

CMS / Sites: Strapi, Payload, Directus, Wagtail, Publii, Astro, Ghost. Email/Marketing: Listmonk, Mautic, Novu, BillionMail-style. Training: Unsloth, Axolotl, LLaMA-Factory, PEFT, DeepSpeed, TRL, Megatron-LM, TorchTune, ColossalAI, litGPT. Orchestration: LiteLLM, Portkey, Semantic Router, OneAPI, LangChain/LlamaIndex/Haystack/Langflow, CrewAI/AutoGen/MetaGPT/Agno/Letta. Many dictated names remain NEEDS VERIFICATION (G-Brain, AlphaClaw, Resun_Robot, Timber, MIKSA/auto-company, specific PewDiePie projects, Cyberger, Quant Trader Lab, H-Compiler, MRG-BENCH, K-A-Pilot, Drift, UseFastLane, Flux Fusion, DUiX variants, exact uncensored community models, etc.).

7. Recommended Models & Tools per Major Tab (Minimum 5–8)

Each tab still has full access to the global model pool. The lists below are the specialized prioritization for that workspace's router and agents. Uncensored/low-guardrail local models are elevated for research, legal, and sovereignty work.

■ LLM / Models (Central Warehouse)

Colibrì (GLM-5.x, Kimi K3, DeepSeek V4 Flash), llama.cpp GGUF pool, Ollama, vLLM/SGLang, OpenRouter free tier, community uncensored variants, vision models (GLM-5.3-Flash etc.), embedding models.

■ Research

Strong reasoning MoEs (Colibrì GLM / DeepSeek / Qwen), long-context models, MinerU + PaddleOCR for documents, SearXNG, Crawl4AI/Firecrawl, Browser Use, citation-aware RAG.

■■ Legal

High-reasoning uncensored/low-guardrail models, MinerU for case documents, structured extraction, evidence library with access control, treaty/government knowledge packs.

■■ OSINT

Browser Use + Playwright, SearXNG, RSSHub/Folo, Firecrawl/Crawl4AI, specialized OSINT sources (OpenCorporates, OpenAlex, Semantic Scholar, ICIJ, Old Maps Online, etc.), local analysis models.

■ Finance / ■ Investments

Reasoning + code models for modeling, financial data tools (interfaces only), no direct ledger. Separate Financial Platform handles custody and rails.

■ Images / Human Studio

ComfyUI + FLUX / SD + IP-Adapter, InvokeAI, reference-conditioned pipelines, character ID + verification layer, diverse skin/body/hair evaluation suite.

■ Video

VACE (Wan2.1-VACE), Wan2.1, LTX ecosystem, Open-Sora, ComfyUI workflows, FFmpeg assembly, talking-head / lip-sync systems, remake engine (shot detection → pose/depth/camera → target character/environment).

■ Music / Audio

Local stem tools, Whisper.cpp, F5-TTS / GPT-SoVITS, generation models, mixing/mastering chain.

■ Coding

OpenHands, Cline, Aider, strong coding models (DeepSeek, Qwen-Coder, Codestral-class, local GGUF), disposable workspaces.

■ Web / Scraping

Browser Use (primary), Playwright, Crawl4AI, Firecrawl, SearXNG, credential-brokered authenticated sessions.

■ Agents / Communications

Agent Creator + library, OpenHands-style + Browser Use + custom, phone/email/messaging integrations (Listmonk, etc.), identity/face for agents where desired.

■ Security / Admin

Keycloak, OpenBao, Wazuh, CrowdSec, Suricata, full audit, threat profiling, automated response playbooks.

8. Agent System & Creator

Goal: rich library of specialized agents (targeting 100+ conceptual agents across categories) plus an Agent Creator so new agents can be defined with task type, allowed workspaces, tools, models, and permission profiles.

Agent Categories (Examples)

- Research agents (deep research, academic, news, document)
- Legal agents (case analysis, claim drafting, precedent retrieval, sovereignty research)
- OSINT agents (person/company investigation, exposure monitoring, social intel)
- Browser / web agents (authenticated browsing, form filling, scraping with approval gates)
- Coding agents (feature implementation, debugging, test writing, PR creation)
- Media agents (image generation with character consistency, video remake, music production)
- Business / lead / CRM agents
- Communications agents (email campaigns via Listmonk etc., messaging, phone where integrated)
- Security agents (monitoring, incident response, report generation)
- Meta agents (orchestration, evaluation, model routing assistance)

Agent Creator Flow

User describes capability → system proposes tools/models/workspaces/permissions → human approves → agent is registered with identity (optional face/voice) → available in library and to orchestration.

Security Rules for Agents

No raw password/credential database access. High-impact actions (financial, account creation, destructive) require explicit human approval via Credential Broker. Every action produces an audit receipt. Agents can propose; humans (or high-assurance policy) dispose.

9. Security, Sandbox & Never-Break Design

Sandbox Architecture

Control Plane + Service Mesh. Major domains (Legal, OSINT, Media, Dev, etc.) run in isolated containers/sandboxes with their own tools, agents, and data mounts. Financial Platform is outside the trust boundary and reachable only via tightly controlled, audited APIs.

Never-Break Plugin Lifecycle

```
NEW MODEL or TOOL
   ↓ Scan (license, malware, permissions)
   ↓ Test in disposable sandbox
   ↓ Benchmark (quality, latency, stability)
   ↓ Security / permission profile
   ↓ Approve & Register
   ↓ Deploy with health checks
   ↓ Continuous monitoring
   ↓ On failure → Automatic Rollback to previous known-good version
```

Threat Response Goals

Detect intrusion attempts, record as much attacker information as possible, generate reports suitable for authorities, and apply blocking/remediation. Continuous monitoring includes the security tools themselves (CrowdSec and others have had advisories).

10. Offline / Hybrid / Online Matrix

Component	Status	Notes / Fallback
Colibrì + local GGUF / llama.cpp	■ Offline	Primary for privacy & huge MoEs
ComfyUI / VACE / Wan local	■ / ■	Local preferred; cloud GPU optional
MinerU / PaddleOCR / Whisper.cpp	■ Offline	Fully local document & audio
Browser Use (local browser)	■ Hybrid	Needs network for target sites
OpenRouter / free APIs	■ Online	Fallback when local unavailable
Kaggle GPU workers	■ Online	Temporary; replaceable by local RTX
Keycloak / Wazuh / local DB	■ Offline	Core control plane stays local
SearXNG / RSSHub	■ Hybrid	Self-hosted but queries external

Fallback chain example: Local primary → Local backup → Free API → Second free API → Paid API. System never hard-stops because one provider is down.

11. Manus 1.6 Implementation Notes & Code Skeleton

Target: produce a reproducible specification and starter package that Manus 1.6 (or a stronger coding agent) can turn into working services. Prefer configuration-driven registration over hard-coded lists.

Recommended Repository Layout

```
KINGDOM-AI-OS/
■■■ frontend/          # Custom Kingdom UI (Open WebUI compatibility layer)
■■■ backend/
■   ■■■ api_gateway/
■   ■■■ auth/          # Keycloak integration
■   ■■■ task_manager/
■   ■■■ agent_manager/
■   ■■■ model_router/
■   ■■■ knowledge/     # RAG + MinerU pipelines
■   ■■■ file_manager/
■   ■■■ audit/
■■■ models/
■   ■■■ local/         # Colibrì, GGUF, etc.
■   ■■■ registry/      # models.yaml
■   ■■■ eval/
■■■ workspaces/        # Per-tab configs & sandboxes
■■■ agents/            # Agent definitions & creator
■■■ integrations/      # OpenRouter, Browser Use, etc.
■■■ infrastructure/    # Docker, K8s, queues, storage
■■■ config/
    ■■■ models.yaml
    ■■■ providers.yaml
    ■■■ routing.yaml
    ■■■ workspaces.yaml
    ■■■ security.yaml
```

Key Configuration Principle

New model or tool → drop into registry YAML → capabilities detected → benchmarked → health-checked → security profile applied → becomes available to router and appropriate workspaces. No source-code edit required for routine additions.

API keys and secrets are never committed. They live in a secrets manager or encrypted local .env and are injected as short-lived scoped credentials by the Credential Broker.

12. Human Studio & Character Consistency (Design Requirement)

This is a first-class subsystem, not an afterthought. The system must produce realistic, diverse human characters across skin tones (including deep brown with accurate melanin response), body proportions (curvy, plus-size, athletic, petite, tall, muscular), facial structures, hair types (straight, curls, coils, locs, braids, protective styles), ages, and presentation, while preserving identity and temporal consistency across images and video.

Technical Approach

- Permanent CHARACTER_ID with face embedding, reference images, body/hair/skin/clothing library, voice, personality, pose/camera references, approved canonical images.
- Pipeline: Prompt → Character Resolver → Reference Pack → Generation → Identity Verification → Temporal Verification → Optional Human Review → Save.
- Conditioning via references, IP-Adapter / adapters, segmentation, pose controls, carefully selected datasets — not a single 'brown skin model'.
- Evaluation suite explicitly tests identity preservation, skin consistency, anatomy (hands/feet), hair, clothing, body consistency, lighting, and temporal stability across multiple skin tones and body types.

13. Immediate Next Steps & Deliverables

This PDF is the Master Architecture Bible (source of truth).

1. Confirm hardware specs (GPU VRAM, system RAM, NVMe size/speed) so Colibrì and local model mapping can be precise.
2. Provide any remaining public API documentation URLs or exact repository links for NEEDS VERIFICATION items (never private keys).
3. Optional: additional UI screenshots or library/image system references.
4. Generate companion ZIP containing: this PDF, starter models.yaml / workspaces.yaml / routing.yaml skeletons, Docker Compose outline, and Manus-oriented prompt package.
5. Begin implementation in stages: Control Plane + Model Registry + one or two pilot workspaces (e.g. Research + Images) → expand.
6. Financial Platform remains a parallel track with its own security boundary.

All names, tools, models, and requirements from the conversation and attachments have been preserved in the master inventory. Verified components are elevated; unverified items remain visible with status flags. The system is designed to be additive forever.

End of Master Architecture Bible — Version 1.0

Kingdom AI OS — Offline-First • Modular • Uncensored Priority • Never Break